The empirical recurrence for OEIS A185986, proved by transfer matrix

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Abstract

OEIS A185986 counts arrays over $\{0, \dots, 1\}$ under a condition comparing a statistic of every 3×3 subblock with those of its neighbours, and carries an empirical recurrence of order 16 contributed by R. H. Hardin. It is true, and it is decidable rather than empirical. The count is a walk count on $S = 10256$ states and is C-finite; whether the conjectured recurrence annihilates it is then settled by a finite exact computation.

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1 The conjecture

OEIS A185986 is "1/128 the number of (n+2)X4 binary arrays with each 3X3 subblock trace equal to some horizontal or vertical neighbor 3X3 subblock trace.". It has offset 1 and begins

$$10, 76, 722, 6036, 52966, 462288, 4038360, 35286678, \dots$$

The entry states, as a conjecture contributed by its author and never marked settled:

> Empirical: a(n)=10*a(n-1)-11*a(n-2)-6*a(n-3)+51*a(n-4)+30*a(n-5)-330*a(n-6)-190*a(n-7)+568*a(n-8)-1888*a(n-9)+3442*a(n-10)-2512*a(n-11)+328*a(n-12)-1862*a(n-13)+4596*a(n-14)-1224*a(n-15)-432*a(n-16)

As of the "Last modified" line on the live entry (Oct 03 2025 03:19:58, revision 9) this is still recorded as empirical, and nothing on the entry records it as proved.

2 What is being counted

Write σ for the sum of its main diagonal, taken on a 3×3 subblock. The contiguous 3×3 subblocks of an $(n + 2) \times 4$ array over $\{0, \dots, 1\}$ form a grid of n rows and 2 columns, and σ labels that grid. The entry requires that every subblock carries the same value of σ as at least ONE of its neighbours horizontally or vertically. The entry counts 1/128 of that number, and the model divides by 128 before anything is compared.

3 The count is a walk count

This condition is NOT a condition on a pair. Whether a subblock has a matching neighbour cannot be decided when its own row is placed: the row BELOW it may yet supply the match. So the state carries, besides the 3 consecutive array rows, a tally for each of the 2 subblocks of the row just completed — a single bit saying whether it has already been matched — counting its matches among its own row and the row above.

A step appends an array row. It fixes the next subblock row, adds the matches between the two rows to the OLD row's tallies, and requires those totals to be acceptable, the old row being final at that moment; the new row's tallies start from its own horizontal matches plus the matches just made. A state is ACCEPTING when its own tally is already acceptable, which is exactly the right condition for the last subblock row, the one with nothing below it. Only the states whose tally came from their own row may BEGIN an array, since the first row has nothing above it.

An array of $n+2$ rows is then a walk of $n-1$ steps and

$$a(n) \;=\; \iota^\top M^{n-1}\tau$$

with ι the indicator of the starting states and τ of the accepting ones, on $S = 10256$ states in all. Getting the two vectors right is the whole of the argument: with ι taken as all-ones the first row would be allowed a match from a row above it that does not exist, and the count would be too large. The entry's own first terms settle it — for the smallest cases the model reproduces every published term.

Over the alphabet $\{0,\dots,1\}$ the statistic takes values in $[0,3]$, so the grid it labels carries at most 4 different labels. At $n=1$ the array has 3 rows and the grid is a single row of 2 subblocks, with only the neighbours inside that row; the entry's first term is 10, which the model reproduces along with every later term, and that is the cheapest check that the grid has been read with the right shape.

In particular a is C-finite. The alignment of the walk index with the entry's own n was checked against its published terms rather than assumed.

4 The proof

Lemma 1. *Let $q(t) = t^{16} - \sum_i c_i t^{16-i}$ be the characteristic polynomial of the conjectured recurrence. The residual $a(n) - \sum_i c_i a(n-i)$ equals $\iota^\top M^j q(M)\tau$ for the corresponding walk index j, so the recurrence holds from some point on if and only if $\iota^\top M^j q(M)\tau = 0$ for all large j, and it is enough to examine $j < S$.*

Proof. Factor M^j out of $M^{j+16} - \sum_i c_i M^{j+16-i}$. For the bound, M^S is an integer combination of $I, M, \dots, M^{S-1}$ by Cayley–Hamilton, so the numbers $u_j = \iota^\top M^j q(M)\tau$ obey the monic recurrence given by the characteristic polynomial of M; S consecutive zeros force every later one, and the last nonzero u_j pins the threshold exactly. ☐

Theorem 2.

$$a(n) \;=\; 10\,a(n{-}1){-}11\,a(n{-}2){-}6\,a(n{-}3){+}51\,a(n{-}4){+}30\,a(n{-}5){-}330\,a(n{-}6){-}190\,a(n{-}7){+}568\,a(n{-}8){-}1888$$

for every $n > 16$.

Proof. The vector $w = q(M)\tau$ was formed by 16 matrix–vector products in exact integer arithmetic and $\iota^\top M^j w$ evaluated for $j = 0, 1, \dots,$ again exactly; those integers vanish from the index corresponding to $n = 16+1$ onwards and the run continued until the working vector was identically zero, which settles every larger j at once. ☐

5 Verification

Three checks, each independent of the algebra above.

First, the model reproduces all 19 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry: the digraph is built by reading the entry's English, and a misreading gives different counts.

Second, the conjectured recurrence was evaluated directly on those published terms, with no matrices involved, and holds wherever the proved range and the data overlap.

Third, the annihilation test of Lemma 1 was carried out in exact integer arithmetic on the whole vector rather than on a sampled prefix.

References

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